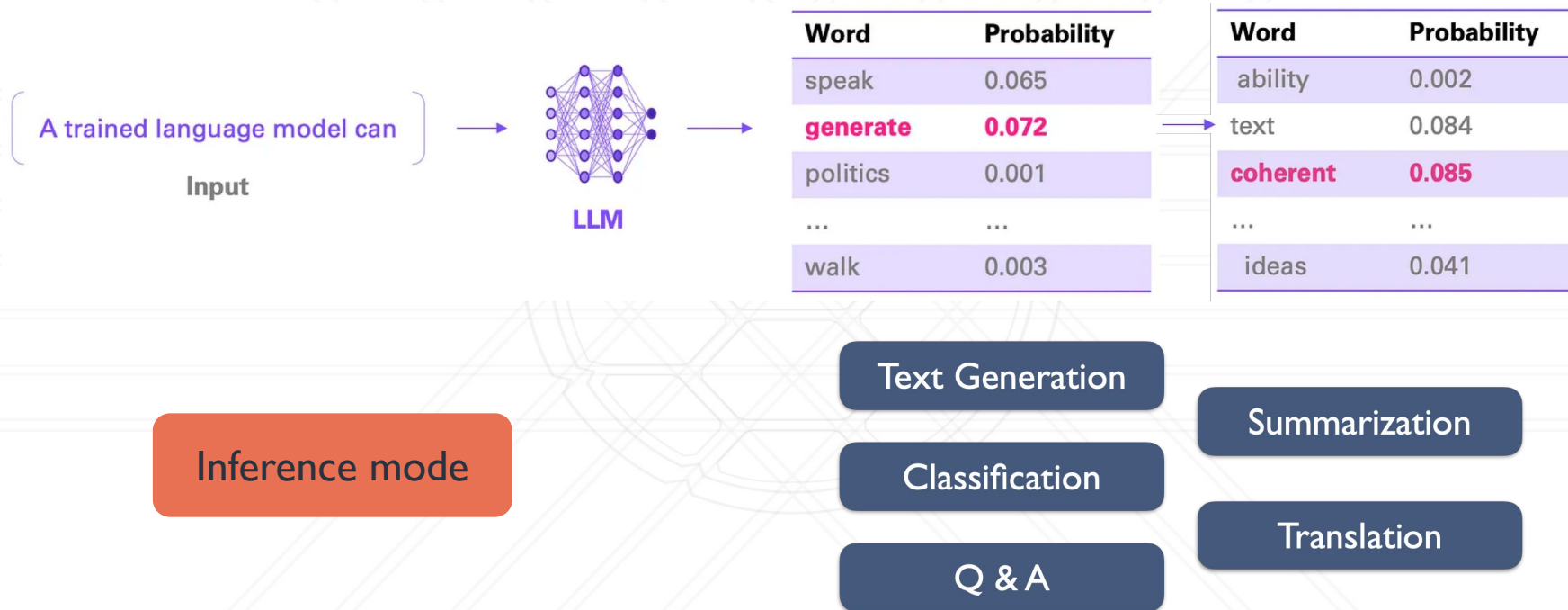


Distributed Deep Learning on GPU-based Clusters

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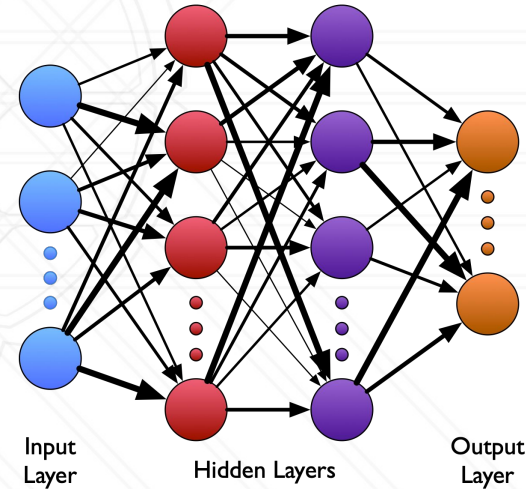
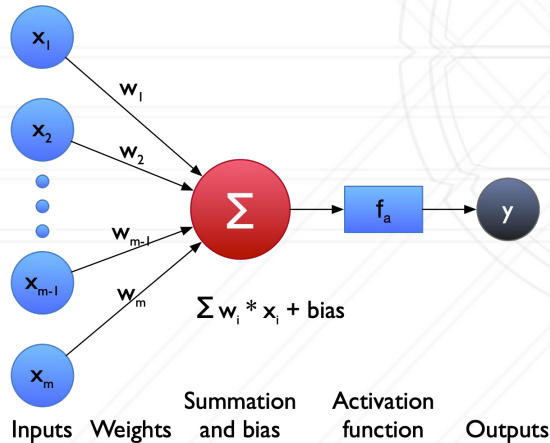
How are neural networks used?



What are neural networks?

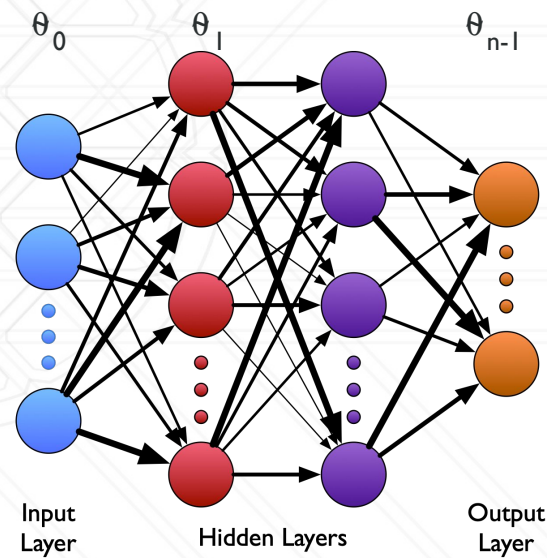
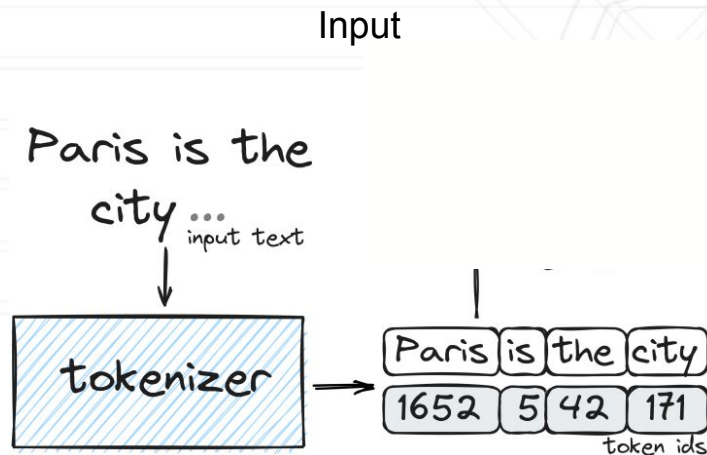
Neural networks are “parametrized” function approximators that can be used to model complex functions

Several layers that process input data



How is a neural network trained?

Problem: Find a set of weights/parameters that best fits the function we are trying to learn over a given training dataset



Other terms and definitions

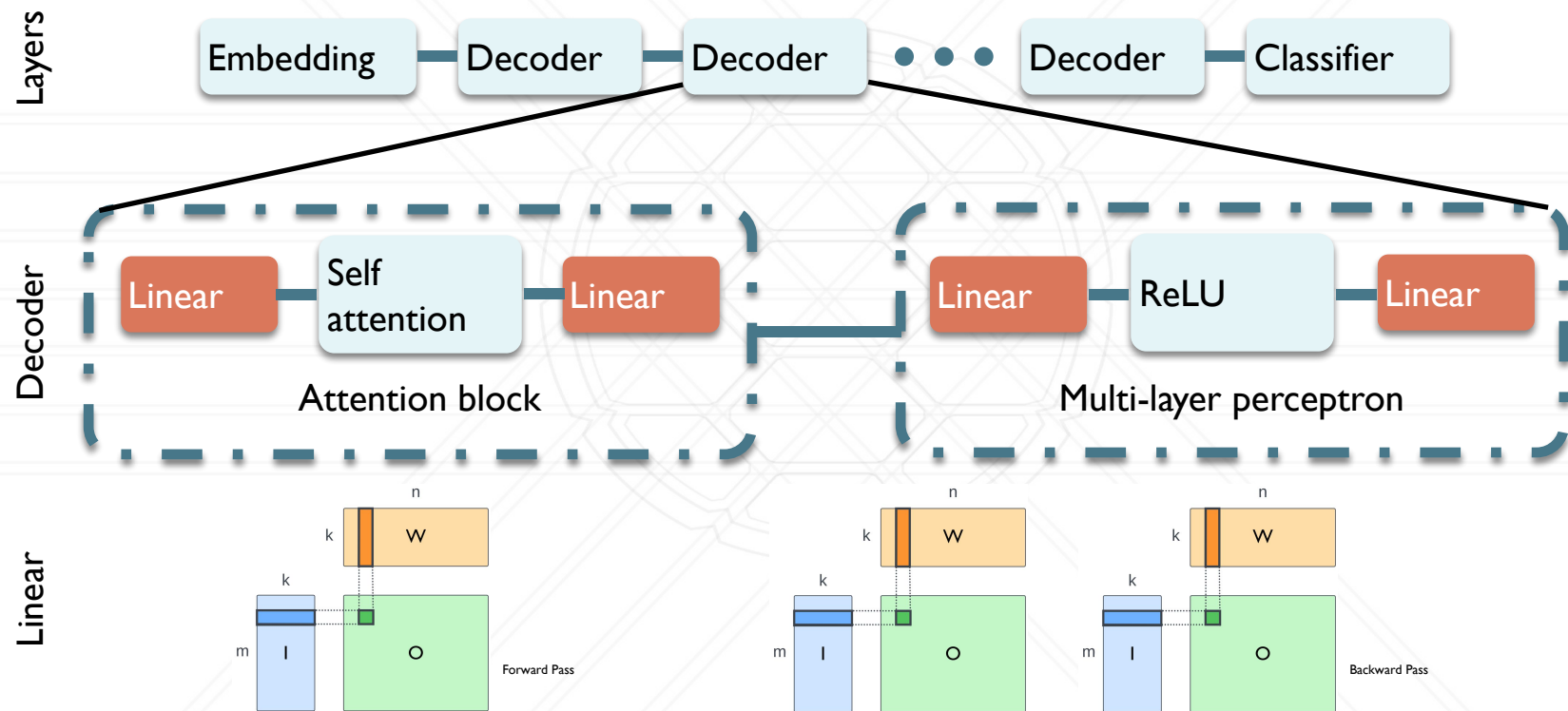
- **Loss:** a scalar proxy that when minimized leads to higher accuracy
- **Learning/training:** task of selecting weights that lead to an accurate function / minimizes the loss
- **Gradient descent:** process of updating the weights using gradients (derivatives) of the loss weighted by a learning rate
- **Batch:** Small subsets of the dataset processed independently
- **Epoch:** One pass over all the batches

- **Fine-tuning:** updating weights for a specific task
- **Inference:** using a trained or fine-tuned model to generate output

Stochastic Gradient Descent

```
while (remaining_batches) {  
    Read a single batch  
  
    Forward pass: perform matrix multiplies to compute  
        output activations, and a loss on the batch  
  
    Backward pass: matrix multiplies to compute gradients of  
        the loss w.r.t. parameters via backpropagation  
  
    Optimizer step: use gradients to update the weights or  
        parameters such that loss is gradually reduced  
}
```

Why is LLM training well-suited for HPC?



Get the tutorial repository

- Clone the git repository as follows:

```
ssh <username>@login.zaratan.umd.edu  
git clone https://github.com/axonn-ai/distrib-dl-tutorial.git
```


PyTorch



- torch – a Python library for tensor computations with GPU support
- torch.nn – implementations of various neural network layers

PyTorch Lightning



- High-level API as an alternative to using PyTorch directly
 - Automatic Mixed Precision
 - Vendor Agnostic - CPU, GPU, TPU
 - **Seamlessly switch between parallel frameworks**
- We will start with looking at single GPU training using PyTorch Lightning

Fine-tuning LLMs

- Fine-tuning is a popular approach to tune pre-trained LLMs for specific tasks
- Instruction fine-tuning (IFT): Train on datasets of [instruction, output] pairs.
- Example use case: build a chatbot by tuning an LLM to follow instructions.

Samples from an IFT dataset

instruction	output
string · lengths	string · lengths
	
Give three tips for staying healthy.	1. Eat a balanced diet and make sure to include plenty of fruits and vegetables. 2. Exercise regularly to keep your body active and strong. 3. Get enough sleep and maintain a consistent sleep schedule.
What are the three primary colors?	The three primary colors are red, blue, and yellow.

These are from the **alpaca** dataset, which we will be using in the tutorial

Using PyTorch Lightning

- Code location in the tutorial repo: train.py

```
CONFIG_FILE=configs/single_gpu.js  
on sbatch train_single.sh
```

Parameter	Value
model_id	microsoft/phi-1_5
dataset_id	alpaca
strategy	single-gpu
global_batch_size	32

Mixed-precision Training

- GPUs have FP32, FP64 and tensor cores
- We can optimize performance by doing some operations in lower precision (16-bit)



Mixed-precision Training

- Code location in the tutorial repo: `train.py`

```
CONFIG_FILE=configs/single_gpu.json sbatch  
train_single.sh
```

Activation Checkpointing

- Activations are outputs of individual layers
- To save memory, we checkpoint only inputs to each layer
 - Regenerate intermediate and output activations as needed in the backward pass
- Code location in the tutorial repo: `train.py`

```
CHECKPOINT_ACTIVATIONS=true sbatch --reservation=sc2024  
run_sequential.sh
```


How to download a different model?

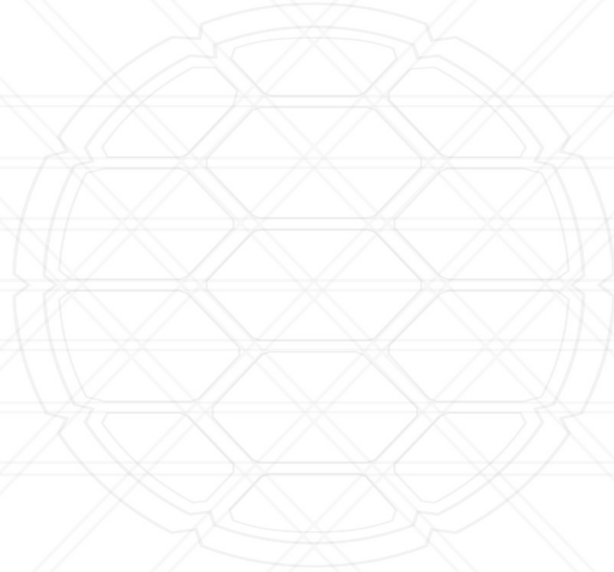
```
cd external; python download.py <model-name>
```

To get a list of available models in LitGPT -

```
python download.py list
```

Where does a model file come from?

- Write your own
- Get a pre-implemented model from somewhere (huggingface, LitGPT, ...)



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